Document Management System

	Sultanate of Oman Ministry of Health The Royal Hospital Department of Pharmacy

Code: PH-SOP-3-Vers.1.2

Effective Date: 01/02/2024
Target Review Date: 31/01/2028

Title: Selected High Alert Medications

Introduction:

High-alert medications are drugs that bear a heightened risk of causing significant patient harm when used in error. Although mistakes may or may not be more common with these drugs, the consequences of an error are clearly more devastating to patients.We hope you will use this list to determine which medications require special safeguards to reduce the risk of errors and minimize harm. This may include strategies like providing mandatory patient education; improving access to information about these drugs; using auxiliary labels and automated alerts; employing automated or independent double checks when necessary; and standardizing the prescribing, storage, dispensing, and administration of these products.

ISMP lists for high alert medication contain two lists which are Acute care setting and Ambulatory care setting. Use these lists to determine which medications require special safeguards at your practice site to reduce the risk of errors.

ISMP List Acute Care Setting:

classes/Categories of Medications

- adrenergic agonists, IV (e.g., EPINEPHrine, phenylephrine, norepinephrine)
- adrenergic antagonists, IV (e.g., propranolol, metoprolol, labetalol)
- anesthetic agents, general, inhaled and IV (e.g., propofol, ketamine)
- antiarrhythmics, IV (e.g., lidocaine, amiodarone)
- antithrombotic agents, including:
  - anticoagulants (e.g., warfarin, low molecular weight heparin, IV unfractionated heparin)
  - Factor Xa inhibitors (e.g., fondaparinux, apixaban, rivaroxaban)
  - direct thrombin inhibitors (e.g., argatroban, bivalirudin, dabigatran etexilate)
  - thrombolytics (e.g., alteplase, reteplase, tenecteplase)
  - glycoprotein IIb/IIIa inhibitors (e.g., eptifibatide)

- dextrose, hypertonic, 20% or greater
- dialysis solutions, peritoneal and hemodialysis
- epidural or intrathecal medications
- hypoglycemics, oral
- inotropic medications, IV (e.g., digoxin, milrinone)
- insulin, subcutaneous and IV
- liposomal forms of drugs (e.g., liposomal amphotericin B) and conventional counterparts (e.g., amphotericin B desoxycholate)
- moderate sedation agents, IV (e.g., dexmedetomidine, midazolam)
- moderate sedation agents, oral, for children (e.g., chloral hydrate)
- narcotics/opioids
- IV
- Transdermal
- oral (including liquid concentrates, immediate and sustained release formulations)
- neuromuscular blocking agents (e.g., succinylcholine, rocuronium, vecuronium)
- parenteral nutrition preparations
- radiocontrast agents, IV
- sterile water for injection, inhalation, and irrigation (excluding pour bottles) in containers of 100 mL or more
- sodium chloride for injection, hypertonic, greater than 0.9% concentration

ISMP List of High-Alert Medications in Community/AmbulatoryHealthcare setting:

Classes/Categories of Medications

- antiretroviral agents (e.g., efavirenz, lamiVUDine, raltegravir, ritonavir, combination antiretroviral products)
- chemotherapeutic agents, oral (excluding hormonal agents) (e.g., cyclophosphamide, mercaptopurine, temozolomide)
- hypoglycemic agents, oral
- immunosuppressant agents (e.g., azaTHIOprine, cycloSPORINE, tacrolimus)
- insulin, all formulations
- opioids, all formulations
- pediatric liquid medications that require measurement
- pregnancy category X drugs (e.g., bosentan, ISOtretinoin)
- Specific Medications
  - carBAMazepine
  - chloral hydrate liquid, for sedation of children
  - heparin, including unfractionated and low molecular weight heparin
  - metformin
  - methotrexate, non-oncologic use
  - midazolam liquid, for sedation of children
  - propylthiouracil
  - warfarin

Policy

- The medication safety practices, special processes and interventions required for the Royal Hospital high alert list must be adopted and implemented in all patient care areas/units of RH facilities.

- Medications used during medical emergencies (e.g. immediate life threatening event) are exempted from the High-Alert Medication Safety Practices in this policy

1. Prescribing
2. Prescription order communications
3. Product labeling
4. Packaging and nomenclature
5. Compounding
6. Dispensing
7. Distribution
8. Administration
9. Education
10. Monitoring

- Orders for High-Alert Medications will include at a minimum:
  1. Patient name and medical record number
  2. Date and time the order is written or entered in the system.
  3. Drug name (generic), dose, route, and date of administration for each drug
  4. Rate and/or duration of administration (if applicable)

1. All elements used to calculate the dose (e.g. height, weight used for medication dosing during that encounter, and/or BSA, if applicable)
2. Allergies

- The following will be available on the patient's medical record:
  - Informed consent (if applicable)
  - All High-Alert Medications will be documented on Medication Administration Record/Anesthesia record/Ambulatory medical record, or other part of the medical record where drug administration is documented

Strategies for the safe use of high-alert medications

Medications may include but are not limited to:

Standardizing high-alert medication concentrations and volume options

- Using pre-mixed solutions (commercially available and pharmacy prepared)
- Using programmable pumps with dosing limits and automated alerts
- Applying warning labels to products as soon as they are received in the pharmacy
- Using visible warning and auxiliary labels according to the organization's policy
- Using patient-specific labelling for unusual concentrations
- Limiting access to high-alert medications in service areas and auditing routinely to assess for items that should .be removed

Segregating and providing directed access to reduce the likelihood of selection errors (e.g., use of automated dispensing cabinets in service areas)

- Providing training about high-alert medications
- Employing redundancies such as automated or independent double checks

ACI Focuses on:

Insulin. Heparin. Narcotics & concentrated electrolytes

SN	Title	Definition	Factors to be verified
1	**Qualified Health Care Practitioner**	Individuals who are qualified to perform double checks or independent double checks within the context of their normal responsibilities and scope of practice. Qualified health care practitioners include physicians, physician assistants, nurse practitioners, certified nurse midwives, registered nurses, and pharmacists.	
2	**Independent Double Check**	A check of the listed factors. It is performed independently by two qualified health care practitioners (MD/RN/Pharmacist), **against thecurrent medication order**, before each high-alert medication is administered. ◦ These checks must be documented on the Medication Administration Record/Anesthesia record/Ambulatory medical record or other part of the medical record where drug administration is documented.	1. Right patient identification using two identifiers – patient name, medical record number. 2. Current patient weight (for weight-based drugs) or body surface area (BSA) for BSA-dosed drugs 3. Right Drug (verified against the current physician order) 4. Right Dose or rate of infusion 5. Right route of administration 6. Right time of administration 7. Pump settings • For patients receiving drugs programmed on the IV pump using the drug library, verify the correct drug, surface area (for BSA dosed drugs), correct dose/rate or duration, and line attachment. The pump settings must be double checked against the current medication order. • For patients receiving drugs on the IV pump that are not built in the drug library, verify the correct drug, concentration, correct entry of patient weight (for weight-based drugs) or body surface area (BSA) for BSA dosed drugs, correct

			factors • Mathematical calculations of the dose and rate for drugs administered I.V will be performed independently by two qualified health care practitioners when: • pumps without “Smart Technology” or drug libraries are used to administer drugs • drugs are not in the I.V. pump drug library; • drugs are ordered during computer downtimes; • These checks must be documented on the Medication Administration Record, Anesthesia record, Ambulatory medical record or other part of the medical record where drug administration is documented.
3	**Time Out**	The period of time **immediately before initiating a high-alert medication administration** /procedure, when two qualified health care practitioners verify the listed factors, at the patient’s side in the location where the medication administration /procedure will be performed. The Time Out must be documented on the medical record at the time of occurrence.	• Availability of any special equipment or special requirements for administration of the medication (e.g. infusion devices), if applicable. For patients with I.V. Pumps – verify the setting and rate of infusion • Correct patient identity – patient name and medical record number. • Correct side and site - verify appropriateness and adequacy of IV access. • Agreement on the medication administration/procedure to be done with the patient. • Correct patient position for epidural and intrathecal medication administration
4	**Pharmacy Pause**	A Pharmacy Pause is defined as checks performed independently by two pharmacists or a pharmacist and a pharmacy technician, before a specified **high-alert medication is dispensed** from the pharmacy	• Right patient identification using two identifiers – patient name, medical record number. • Right Drug (verified against original physician order)

5	**Hand-over**	specific information from one caregiver to another for the purpose of ensuring the continuity and safety of the patient's care. Hand-over occurs when a nurse transfers responsibility for the patient for the remainder of the workday – e.g. change of shift. Hand-over does not include coverage for breaks or meal periods	

Special procedure for different medical agents or route of administration to maximize safety

- **Medications administered via the Intrathecal Route**

An independent double check shall be conducted **in the Pharmacy** by two health care professionals (e.g. two pharmacists, one pharmacist and one pharmacy technician, one pharmacist and one physician) after the preparation of the intrathecal dose to assure it is prepared and labeled correctly.

For intrathecal drugs prepared in a sterile environment **outside the Pharmacy** (e.g., operating room, labor and delivery), the independent double check shall be conducted by any two health care professionals (e.g., two registered nurses, one physician assistant and one nurse anesthetist, one certified nurse midwife and one physician, etc.) after the preparation of the intrathecal dose to assure it is prepared and labeled correctly

- Label should include the warning: "Caution: For intrathecal use only."
- Independent Double Check and Time Out Required.
- Intrathecal chemo medication will be compounded and dispensed in a syringe that is *10 mL or smaller. No other cytotoxic drugs will be present at the patient bedside during the intrathecal chemo administration process.*
- A "time out," including an independent double check, shall be conducted at the bedside immediately prior to the administration of all doses of intrathecal medications. (See definition above.) These checks shall be documented in the medical record

- **Continuous intravenous infusions of Heparin and Argatroban**

- The abbreviation "u" will not be accepted in heparin medication orders. Units must be spelled out.
- A standard concentration will be utilized for all continuous heparin and argatroban infusions.
  - Heparin 50 units/mL
  - Argatroban 250mg in 250mL
- order sets shall be utilized for prescribing continuous infusions of heparin.
- Infusion pumps will be used with the safety software activated when heparin and argatroban are administered.

- **Continuous intravenous infusions of Insulin**

- The abbreviation "u" shall not be accepted in the medication order. Units must be spelled out.
- A standard insulin concentration of 1 unit/ml shall be utilized for all adult continuous insulin infusions.
- Prime insulin tubing with 20 mL.
- Maximum will be insulin 250 units in 250 mL.
- Order sets shall be utilized for prescribing continuous infusions of insulin.
- Infusion pumps will be used with the safety software activated when insulin drips are administered.

- Neuromuscular blockers shall only be stored in specific areas within the hospital, e.g. OR, , Critical Care (PICU/NICU/ICU), ED, Cath Lab.
- Distinctive labeling and/or storage shall be utilized to distinguish neuromuscular blockers from other medications outside the O.R., e.g. segregation, colored bins, etc. Pharmacy will affix a label to all vials prior to dispensing to areas outside the OR – e.g. Critical Care, ED, PACU
- All infusions of neuromuscular blockers shall be affixed with the following label prior to being dispensed from Pharmacy:Caution: “Paralyzing Agent” Patient must be on a ventilator or other ventilation support
- An independent double check by two qualified health care practitioners (e.g. nurse, pharmacist, physician) is required for each intravenous push dose. For infusions, independent double checks are required initially, at each bag change, rate change, and at hand off
- Infusion pumps will be used with the safety software activated when neuromuscular blocker infusions are administered. When a drug is not in the pump library, the pump will be programmed by entering the necessary fields of data based upon the mode being used, and independent calculations of the dose and rate must be performed prior to administration.
- Policies and procedures regarding the responsibilities of anesthesia providers in the High Alert Medication Program are specified in Regional High Alert Medication Safety Practices for Anesthesia.
- A “time out,” shall be conducted at the bedside immediately prior to the administration of all bolus doses and infusions of neuromuscular blocking agents. These checks shall be documented in the medical record.
- Whenever feasible, Sets/preprinted orders should be utilized for prescribing neuromuscular blocking agents. Orders must also state: “Patient must be on a ventilator”

- **Opiate/Narcotic infusions including PCA therapy**

- Whenever feasible, Order Sets/preprinted orders should be utilized for prescribing opiate/narcotic infusions and PCA therapy.
- The following standard concentrations shall be utilized for PCA therapy:
  - morphine 1 mg/mL
  - fentanyl 10 mcg/mL
- In clinical situations where more concentrated infusions are required, the syringes/bags shall be affixed with a “Non Standard Concentration” label

- **All medication infusions administered via the epidural route including Opiate/Narcotic Medications**

- Whenever feasible, preprinted orders should be utilized for prescribing epidural infusion.
- All epidural infusions shall be administered utilizing a programmable/advanced mode pump and label on pump display with appropriate medication.
- Whenever feasible, commercially prepared bags of medications shall be utilized for epidural infusion.
- Specific identified (e.g. yellow stripe) tubing without injection ports shall be utilized for administering epidural infusions.
- Infusion pumps will be used with the safety software activated when specific drugs, concentrations and volumes are in the pump library. When a drug is not in the pump library, the pump will be programmed by entering the necessary fields of data based upon the mode being used, and independent calculations of the dose and rate must be performed prior to administration.

- **Intravenous, Intraperitoneal, Intraarterial, Intrahepatic and Intrapleural Cytotoxic Chemotherapy**

- Verbal orders shall **not** be accepted when prescribing intravenous, intraperitoneal, intra-arterial, intrahepatic and intrapleural cytotoxic chemotherapy with the exception of date or time changes and clarifications.

- All elements used to calculate the initial dose or change of treatment of a chemotherapy agent should be included on the order or prescription (height, weight, and/or BSA if applicable)
- Indication that written informed consent was obtained for research protocols
- Allergies
- Chemotherapy agent name, dose, route, and date of administration for each drug
- Cycle number and/or week number as appropriate to the regimen, if applicable
- All doses of intravenous, intraperitoneal, intraarterial, intrahepatic and intrapleural cytotoxic chemotherapy shall be affixed with a “Caution: Chemotherapeutic Agent” label.
- Missing dose requests for intravenous, intraperitoneal, intra-arterial, intrahepatic and intrapleural cytotoxic chemotherapy shall be investigated immediately by a pharmacist and a replacement dose shall not be dispensed until the disposition of the first dose is verified

- **Intravenous Infusions of Ketamine, Phenobarbitone, propofol, midazolam, lorazapam, dexmedetomidine (Precedex)**

- Whenever feasible, Order Sets/preprinted orders should be utilized for prescribing infusions of ketamine,phenobarbitone,propofol, midazolam, and lorazepam, and dexmedetomidine (Precedex).
- An independent double check by two qualified health care practitioners (e.g. nurse, pharmacist, physician) is required initially, at each bag change, rate change, and at hand-over.
- Infusion pumps will be used with the safety software activated when specific drugs, concentrations and volumes are in the pump library.
- When a drug is not in the pump library, the pump will be programmed by entering the necessary fields of data based upon the mode being used and independent calculations of the dose and rate must be performed prior to administration.

- **Medication Administration to Pediatric and Neonatal Patients Special processes to maximize safety**

- The High Alert Medications for use in all Pediatric and Neonates will include those drugs and medication management requirements in the adult High Alert Medication Policy.
- In addition, an independent double check by two qualified health care practitioners (e.g. nurse, pharmacist, physician) is required initially, at each bag or syringe change, and at hand-offs for these medications:
  - Remicade, all routes
  - Chloral hydrate, all routes
  - Insulin, all routes
  - Digoxin oral and I.V.

- **DOPamine, DOBUTamine, Epinephrine, Norepinephrine and Phenylephrine Infusions**

- An independent double check by two qualified health care practitioners (e.g. nurse, pharmacist, physician) is required initially, at each bag change, rate change, and at hand-offs.
- Infusion pumps will be used with the safety software activated when specific drugs, concentrations and volumes are in the pump library.
- When a drug is not in the pump library, the pump will be programmed by entering the necessary fields of data based upon the mode being used
- independent calculations of the dose and rate must be performed prior to administration.

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Document Management System

	Sultanate of Oman Ministry of Health The Royal Hospital Department of Pharmacy

Code: PH-POL-6-Vers.1.0

Effective Date: 31/01/2023
Target Review Date: 02/02/2025

Title: Management of High Alert Medications: Independent Double Check (IDC

Purpose

TheRoyal Hospital's policy is to promote safety by avoiding preventable injuries associated with high alert medications (management of high alert medication spolicy). Independent double check by a second healthcare provider prior to administration and at handoff of care for the purposes of safety and accuracyis required. The policy outlines the process and documentation related to independent double check (IDC).

Intra-operative or procedural medications administered by anesthesia and emergency administration of drugs fall outsidethe scope of this policy.

Policy

A medication independent double check is required:
- Prior to administration of selected high-risk medications
- At the time of shift change or any handoff of care

High Alert medications requiring an independent double check will be designated assuch on the electronic Medication Administration Record (eMAR). Documentation of the independent double check will be entered in the eMAR.

The **first check** must be conducted by a provider licensed to prescribe, dispense, oradminister medications. This includes, but is not limited to, nurses,pharmacists, physicians and licensed independent practitioners.

The **second check** will be conducted by a licensed provider, technologist, or technician qualified to complete an independent double check. A technician will be deemed qualified to complete the double check once he/she has completed a written and/or observational competency validation covering all medications that thetechnician may be asked to independently double check.

Ther equirement for the high alert medication independent double check or secondprovider verification for insulin:

1. with each dose/injection (Insulin)
2. for infusions:

a. at the time of initiation of therapy
b. at the time of a concentration change
c. at the time of a bolus dose
e. with any dose change
f. with any bag change
g. as part of the change of each shift or any handoff of care

1. For initial dose or initiation of a new infusion:

A. The healthcare provider preparing toadminister medication will prepare the medication and prepare/retain the following items for use by a second provider who will double check themedication preparation:

1) The original medication package with labeling intact or vial from which the medication was drawn.
2) The eMedication Administration Record.

3) The prepared medication ready for administration with labeling intact.

B. A second provider will assure the following:

1) The medication is prepared according to the prescription

2) The medication prepared for administration matches the five rights.

i. right medication
ii. right dose or rate, including independent double check of any calculationsand verification of pump settings

iii. right route, including line reconciliation
iv. right frequency
v. right patient

3) Insome instances, the medication packaging or vial must also be present to checkthat the prepared medication is correct, e.g., insulin doses

4) Once the second provider performs the independent double check and both providersare satisfied that the medication is accurate, the independent double checkwill be entered on the electronic patient medical records (ePMR) or in the medication administration record. For areas that do not document on the ePMR,the independent double check will be recorded on the same document wheremedication administration is recorded.

5) The independent double check will be conducted PRIOR to the medication being administered.

2. For shift change or any handoff of care:

A. The receiving provider will assure the following:

i. right medication
ii. right dose or rate and verification of pump settings

iii. right route, including line reconciliation
iv. right frequency
v. right patient

B. Once the safety check is complete and the nurse is satisfied that the medication(s)are accurate, the independent double check will be entered in the electronic record or manually documented in the hardcopy.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
ISMP's List of High-Alert Medications. www.ismp.org/highalertmedications.pdf	Institute for safe medication practices	2012	

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Review Details

	Review BY	Review Date
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